

Process Efficiency as a Barrier to Innovation

Online Appendix

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This Online Appendix accompanies the paper. Section, equation, table, figure, proposition, and footnote numbers without an appendix letter refer to the main text. Appendix A documents how subsidies and patents are distributed across firm sizes, B collects the production-side algebra, C the proofs, and D the dynamic choice of process efficiency. Appendix E constructs the breakthrough-patent measure, F reports robustness checks for the reduced-form results, G describes the standard errors, and H reports supplementary estimation results, including the restricted model, the crowding-out coefficient in the model, and the calibrated economy. Appendix I describes the computational algorithm, J the robustness of the quantitative results, and K an extension with heterogeneous firm types.

A. THE ALLOCATION OF SUBSIDIES AND PATENTS ACROSS FIRM SIZE

This appendix documents how innovation subsidies and patents are distributed across firm sizes in China. I group firms into the ten employment-size bins of the Economic Census classification of the National Bureau of Statistics. Since some industries are both larger and more subsidy-intensive than others, I compute all comparisons within 4-digit industry $\times$ year cells with at least 10 firms, which leaves 1,889 industry-years. I also reweight firms so that each size bin carries equal weight, which does not affect the subsidy yields. The two patterns below are unchanged, in results not reported here, when I instead reweight the stratified CTS sample to the bin shares of the National Economic Census or restrict the sample to 2016–2020.

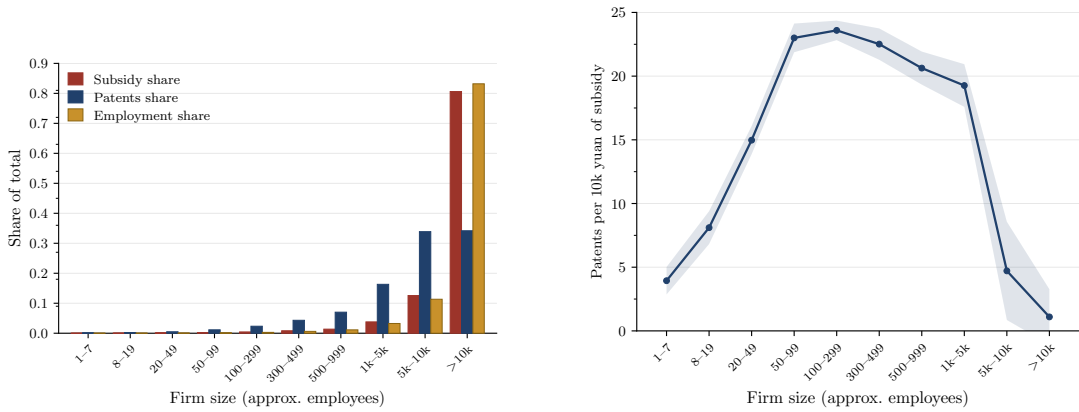
A.1. *Subsidy and patent shares*

Panel (a) of Figure A.1 shows the share of subsidies, patents, and employment in each size bin. Firms with more than 10,000 employees account for 81% of subsidies and 83% of employment but only 34% of patents. In the middle of the size distribution, patent shares exceed subsidy shares. Firms with 1,000–5,000 employees, for example, account for 16% of patents and 4% of subsidies. The sharpest break is between the 5,000–10,000 bin and the bin above 10,000 employees. The subsidy share jumps from 13% to 81% and the employment share from 11% to 83%, while the patent share remains

at about 34%. Hence subsidies track employment across firm sizes, and both diverge from patents.

A.2. *Patents per subsidy yuan*

I define the *subsidy yield* as the number of patents per 10,000 yuan of subsidy, computed among the firms that receive subsidies in each cell.¹ Panel (b) of Figure A.1 shows that the yield is hump-shaped in firm size. It rises from about 4 patents per 10,000 yuan at the smallest firms to a peak near 24 at 100–200 employees, stays high through the middle of the distribution, and falls toward zero above 3,000 employees. The 95% confidence bands at the peak and in the right tail do not overlap.



(a) Distribution of subsidies, patents, and employment across firm size.

(b) Patents per 10,000 yuan of subsidy by firm size.

FIGURE A.1
Subsidy allocation and yield across firm size

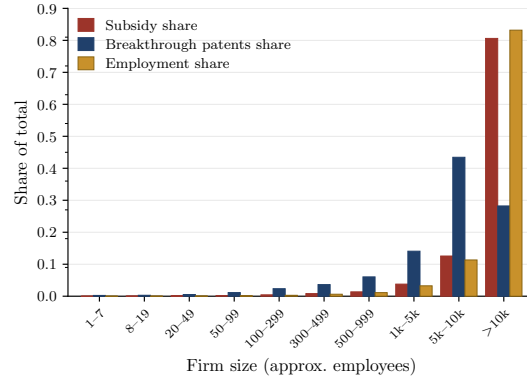
Notes: Panel (a) shows the shares of subsidies, patents, and employment accruing to each NBS employment-size bin when firms are reweighted so that every bin carries equal weight. Panel (b) shows patents per 10,000 yuan of subsidy among recipients, with the 95% confidence band computed from between-cell variation; the yield profile is invariant to the reweighting. Yields are winsorized at the 99th percentile. Bin-level statistics require at least three subsidized firms per cell and five contributing cells per bin.

A.3. *Robustness to strategic patenting*

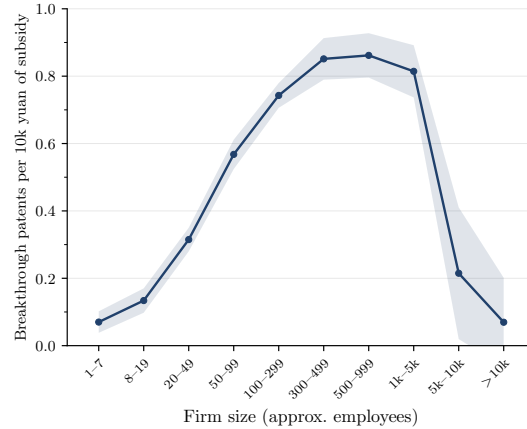
Chinese firms have well-documented incentives to file low-quality patents in order to qualify for High-Technology Enterprise status, tax preferences, and procurement eligibility. If small and mid-sized firms file for these reasons more often than large firms, the patterns above could reflect filing behavior rather than innovation. I therefore reconstruct both panels of Figure A.1 using only breakthrough patents, the top 10%

1. The yield is defined only for recipients. If the small firms that receive subsidies are more capable than the typical small firm, the hump could partly reflect selection rather than the return to a marginal yuan of subsidy. The regressions in Section 3.2 do not have this problem, since they use all firms and exploit variation in subsidy concentration within industries.

of patents by the inventiveness measure of Appendix E. Figure A.2 shows that both patterns remain. Firms with more than 10,000 employees account for 81% of subsidies but only 28% of breakthrough patents, which are concentrated in the 1,000–5,000 and 5,000–10,000 bins (14% and 43%). Breakthrough patents per yuan of subsidy have the same hump shape, with the highest yields at 200–2,500 employees and yields close to zero above 5,000. Hence the mismatch between where subsidies go and where patents come from is not driven by low-quality filings.



(a) Distribution of subsidies, breakthrough patents, and employment.



(b) Breakthrough patents per 10,000 yuan of subsidy.

FIGURE A.2

Subsidy allocation and breakthrough patent yields across firm size

Notes: Breakthrough patents are the top decile of year-demeaned log inventiveness (Appendix E). In panel (a), each bar shows the share of subsidies, breakthrough patents, or employment accounted for by firms in that size bin when the bins are reweighted to equal firm weight. Panel (b) reports yields among subsidy recipients, winsorized at the 99th percentile before aggregation, with the shaded 95% confidence band. All statistics are computed within 4-digit industry $\times$ year cells.

B. PRODUCTION-SIDE ALGEBRA

This appendix derives the profit coefficient $\tilde{p}$ of Section 2.1. Given prices $\{p_j\}$, the final-good producer solves

$$\max_{\{k_j\}} L^\beta \int_{j \in \mathcal{N}} z_j^\beta k_j^{1-\beta} dj - \int_{j \in \mathcal{N}} p_j k_j dj,$$

with first-order condition

$$k_j = (1 - \beta)^{1/\beta} L z_j p_j^{-1/\beta}.$$

Demand has constant elasticity $1/\beta$ and every line has marginal cost $w_u/\bar{q}$, so the leader charges $p_j = w_u/[(1 - \beta)\bar{q}]$ on every active line. Substituting,

$$k_j = (1 - \beta)^{2/\beta} L z_j \bar{q}^{1/\beta} w_u^{-1/\beta}, \quad \Pi_j = \frac{\beta}{1 - \beta} \frac{w_u}{\bar{q}} k_j = \beta (1 - \beta)^{\frac{2}{\beta}-1} L z_j \bar{q}^{\frac{1}{\beta}-1} w_u^{1-\frac{1}{\beta}}.$$

Since $z_j^\beta k_j^{1-\beta} \propto z_j$, aggregate output is $Y = (1 - \beta)^{2(1-\beta)/\beta} L \bar{q}^{(1-\beta)/\beta} w_u^{-(1-\beta)/\beta} Z$, where $Z \equiv \int_{j \in \mathcal{N}} z_j dj$. The first-order condition for assembly labor, $w_u = \beta Y/L$, then gives the wage

$$w_u = \beta^\beta (1 - \beta)^{2(1-\beta)} \bar{q}^{1-\beta} Z^\beta.$$

Substituting back, the powers of $1 - \beta$ collect to $\frac{2-2(1-\beta)^2}{\beta} - 1 = 3 - 2\beta$ and those of $\bar{q}$ to $\frac{1-(1-\beta)^2}{\beta} - 1 = 1 - \beta$, so that

$$\Pi_j = \beta^\beta (1 - \beta)^{3-2\beta} L z_j \bar{q}^{1-\beta} Z^{\beta-1}. \quad (\text{B.1})$$

Line profit (B.1) is linear in $z_j = q_j(1 + \psi_f)$. Hence normalized line profit is

$$\frac{\Pi_j}{\bar{q}} = \tilde{p} x_j (1 + \psi_f), \quad \tilde{p} = \beta^\beta (1 - \beta)^{3-2\beta} L (Z/\bar{q})^{\beta-1},$$

and summing over the lines of a firm gives operating profit $\tilde{p}(1 + \psi_f) Q_f$. Since $\bar{q}$ is the average quality of active lines, $Z = \bar{q} \int_{\mathcal{N}} x_j (1 + \psi_{f(j)}) dj = \bar{q} \Phi_A \bar{\Psi}$, where

$$\bar{\Psi} \equiv \int x (1 + \psi') dS(x, \psi')$$

is the quality-weighted average of $1 + \psi$ across active lines. Therefore

$$\tilde{p} = \beta^\beta (1 - \beta)^{3-2\beta} L (\Phi_A \bar{\Psi})^{\beta-1}, \quad (\text{B.2})$$

Since $\beta < 1$, higher average process efficiency lowers $\tilde{p}$.

Two further results are used in the main text. First, output is $Y = \beta^{\beta-1} (1 - \beta)^{2(1-\beta)} L \bar{q} (\Phi_A \bar{\Psi})^\beta$, so Y and $\bar{q}$ grow at the same rate on the balanced growth path. Second, unskilled labor in intermediate production is $\int_{\mathcal{N}} k_j / \bar{q} dj = (1 - \beta)^2 L / \beta$, a constant multiple of assembly labor, as footnote 2 states.

C. PROOFS

This appendix proves Propositions 1–4 and derives the growth rate (19) and the general-equilibrium effects of a subsidy mentioned in Section 2.6. The notation is that of Section 2. In addition, I write

$$\mathcal{L}f(p) \equiv \sum_j [f(p) - f(p \ominus x_j)]$$

for the value a firm loses when its lines are displaced, as in (14).

Two lemmas

LEMMA C.1. *Let a defender with process efficiency ψ face challengers i with attempt intensities $a_i \geq 0$ and process efficiencies $\psi_i \geq 0$, so that each of its lines is displaced at rate*

$$T(\psi) = \sum_i a_i \Lambda((1 + \lambda)(1 + \psi_i) - (1 + \psi)).$$

Then T is decreasing in ψ , and $T(\psi) \rightarrow 0$ as $\psi \rightarrow \infty$ whenever $\sum_i a_i < \infty$. In the step-function limit $\kappa \rightarrow \infty$, a challenger with $\psi_i = 0$ contributes zero whenever $\psi > \lambda$.

Proof. A higher ψ lowers the challenger's advantage in every term. Since Λ is increasing, every term falls, hence T falls. As $\psi \rightarrow \infty$, each term goes to zero, and since the terms are bounded by a_i with $\sum_i a_i < \infty$, $T(\psi) \rightarrow 0$ by dominated convergence. The argument is the same when the sum is an integral over challengers. Finally, a challenger with $\psi_i = 0$ has advantage $\lambda - \psi$, and as $\kappa \rightarrow \infty$, $\Lambda(\lambda - \psi) \rightarrow \mathbf{1}\{\lambda > \psi\}$ away from $\psi = \lambda$, which is zero when $\psi > \lambda$. $\square$

LEMMA C.2. *Suppose the subsidy rate $\sigma(p)$ is weakly increasing in size, which includes the case without subsidies. Then $\hat{V}(p \oplus y) > \hat{V}(p)$ for every portfolio p and every $y > 0$.*

Proof. The idea is that a firm with an extra line can copy the R&D of a firm without it and do at least as well. Let firm a start from $p \oplus y$ and firm b from p , and let a choose the same attempt rates as b at every date. Put the two firms on the same probability space: their attempts arrive at the same times and land on the same lines, and one uniform draw decides whether each attempt succeeds. Since a is at least as large as b and its subsidy rate is at least as high, its process efficiency is at least as high, so a succeeds whenever b does. For the same reason, a loses each line that both firms hold at a lower rate (Lemma C.1), so the draws can be arranged so that a loses such a line only when b loses it too. The two firms exit together, and their relative qualities fall at the same rate. The lines of b are therefore always a subset of the lines of a . Hence a earns at least the flow profit of b at every date, and strictly more as long as it holds a line that b does not, while its R&D cost $(1 - \sigma_a)w_s h_X$ is no larger. Copying b thus gives a a value above $\hat{V}(p)$, and its optimal value $\hat{V}(p \oplus y)$ is at least as high. $\square$

Proof of Proposition 1 (Process efficiency insulates large incumbents).

Without subsidies, $\psi(p) = \psi^*(Q(p))$. The displacement rate (12) is a special case of T in Lemma C.1. The challengers are entrants, with intensity X_e and $\psi = 0$, and incumbents with portfolio p'' , with intensity $X^*(p'') dM(p'')$ and process efficiency $\psi(p'')$. Their total intensity is the aggregate attempt rate $\bar{\tau}$, which is finite. Since $\psi^*(Q)$ rises without bound in Q by (5), the lemma implies that $\tau(\psi^*(Q))$ is decreasing in Q and converges to zero. For $Q > \bar{Q}_e$, (10) gives $\psi^*(Q) > \lambda$, so in the step-function limit no entrant can displace the firm.² An incumbent challenger with process efficiency ψ'' displaces it if and only if $(1 + \lambda)(1 + \psi'') > 1 + \psi^*(Q)$. The required ψ'' rises with Q , so the set of challengers that qualify shrinks as Q grows. $\square$

Proof of Proposition 2 (Larger firms devote more R&D labor to process efficiency).

Without subsidies, process R&D labor grows exactly like Q^η , since $h_\psi(p) = c_\psi[\psi^*(Q)]^\omega \propto Q^\eta$ by (6). Product R&D labor grows with the value of an extra line. If firm value grows like Q^η , an extra line is worth about $Q^{\eta-1}$, and product R&D labor grows like $Q^{(\eta-1)/\gamma}$, which is slower than Q^η exactly when $\omega > 1/\gamma$. The work lies in showing that firm value does grow like Q^η .

To this end, I conjecture a solution of (14) whose value is asymptotically a power of aggregate quality, $\hat{V}(p) = V_0 Q(p)^b (1 + o(1))$ as $Q(p) \rightarrow \infty$, for some $b \geq 1$ and $V_0 > 0$, with increments and derivatives of the corresponding order. To find b , I compute the order of each term in (14).

Gain from an attempt. For $b \geq 1$ and $y \geq 0$,

$$(Q + y)^b - Q^b \leq b(Q + y)^{b-1}y \leq b2^{b-1}(Q^{b-1}y + y^b).$$

The new line has relative quality $y = 1 + \lambda$ when an inactive line re-enters the economy and $y = (1 + \lambda)x$ after a capture. Since $\int x dS = 1$ and $\int x^\eta dS < \infty$ by condition (vi) of Definition 1, $G(p) = O(Q^{b-1})$ at $b = \eta$. By (15),

$$h_X(p) = \left[\frac{(1 - \gamma) \theta G(p)}{w_s} \right]^{1/\gamma} = O(Q^{(b-1)/\gamma}),$$

and the R&D surplus at the optimum, $\gamma X^* G = \frac{\gamma}{1-\gamma} w_s h_X$, is of the same order.

Loss of lines. For $b \geq 1$, $Q^b - (Q - x_j)^b \leq b Q^{b-1} x_j$, so

$$0 \leq \mathcal{L}\hat{V}(p) \leq b V_0 Q^{b-1} \sum_j x_j (1 + o(1)) = b V_0 Q^b (1 + o(1)),$$

regardless of the composition of p . Since $\tau(\psi^*(Q)) \rightarrow 0$ by Proposition 1, the loss term in (14) is $o(Q^b)$.

Decline in relative quality. $\sum_j x_j \partial \hat{V} / \partial x_j = b V_0 Q^b (1 + o(1))$.

2. At $Q = \bar{Q}_e$ the entrant's advantage is zero and $\Lambda(0) = 1/2$, so the threshold is sharp only in the limit.

Matching orders. The left-hand side of (14) is of order Q^b . On the right-hand side, flow profit is of order Q^η , the R&D surplus of order $Q^{(b-1)/\gamma}$, and the remaining terms of order at most Q^b . Consistency therefore requires

$$b = \max \left\{ \eta, \frac{b-1}{\gamma} \right\},$$

which has two roots, $b = \eta$ and $b = 1/(1 - \gamma)$.

Assumption 2 is equivalent to $\eta < 1/(1 - \gamma)$, since $\eta = \omega/(\omega - 1)$ is decreasing in ω and equals $1/(1 - \gamma)$ at $\omega = 1/\gamma$. Both roots are therefore possible, and the second is larger. At $b = 1/(1 - \gamma)$, the first-order condition gives $h_X = O(Q^{1/(1-\gamma)})$, which grows faster than flow profit, so the flow payoff $\pi - w_s h_X$ diverges to $-\infty$. Firm value is then positive and growing while current payoffs fall without bound, sustained only by the anticipation of further growth. The transversality condition $\lim_{T \rightarrow \infty} e^{-\hat{r}T} \mathbb{E}[\hat{V}(p_T)] = 0$ rules out this branch.

On the remaining root $b = \eta$, Assumption 2 implies $(\eta - 1)/\gamma < \eta$, so the R&D surplus is of lower order than flow profit. Matching the terms of order Q^η gives $\hat{r} V_0 = \hat{D} - g \eta V_0$, namely,

$$V_0 = \frac{\hat{D}}{\hat{r} + \eta g}.$$

Note that V_0 does not depend on how quality is spread across the lines of the firm, as the conjecture requires. The reason is that a large firm almost never loses a line, so which line it would lose does not matter.

Hence $h_X(p) = O(Q^{(\eta-1)/\gamma}) = O(Q^{1/[\gamma(\omega-1)]})$, while $h_\psi(p) \propto Q^\eta$. Comparing exponents,

$$\eta > \frac{1}{\gamma(\omega - 1)} \iff \frac{\omega}{\omega - 1} > \frac{1}{\gamma(\omega - 1)} \iff \omega > \frac{1}{\gamma},$$

so under Assumption 2, $h_X(p)/h_\psi(p) \rightarrow 0$ and $\chi(p) \rightarrow 1$, regardless of the composition of the firm's portfolio. $\square$

The limit does not establish that χ is monotone in Q , since the ratio h_X/h_ψ contains lower-order terms that need not vanish monotonically.

On condition (vi). Each line goes through a simple cycle. While it is active, its relative quality falls at rate g as average quality grows, and it jumps by the factor $1 + \lambda$ whenever the line is improved. When its owner exits, at rate φ , the line becomes inactive, and it later re-enters the economy at quality $(1 + \lambda)\bar{q}$, close to the economy-wide mean. Exit thus pulls every line back toward the mean, and this keeps the distribution of relative quality from spreading out forever. Since exit is rare relative to improvement, however, the right tail is thick. Suppose, for example, that every active line is displaced, and hence improved, at a common rate ν . The moment of order k of relative quality is then finite if and only if

$$\nu[(1 + \lambda)^k - 1] - g k < \varphi,$$

where $g = \lambda(\nu + \varphi)$ by (19). At the calibrated values $\lambda = 1.08$, $\varphi = 0.004$, and $g = 0.023$, the largest such k is about 1.7. This is above $\eta \approx 1.44$, so condition (vi) holds, though not by a wide margin. The tail is thinner still if high-quality lines tend to be held by firms with high process efficiency, since those lines are displaced, and hence improved, less often.

Proof of Proposition 3 (Subsidies to large firms fund process efficiency).

The proof uses the idea that a subsidy scales each type of R&D by a fixed factor, which means that the firm puts the extra labor in the activity to which it already allocates most of its R&D. Consider a marginal subsidy to a firm with portfolio p , which lowers its effective skilled wage to $(1 - \sigma)w_s$. The firm is atomistic, so it takes w_s , its success probabilities, and $\hat{V}(\cdot)$ as given.

Process efficiency solves (4) at the subsidized wage, so $\psi^*(Q; \sigma) = (1 - \sigma)^{-1/(\omega-1)}\psi^*(Q)$ and

$$h_\psi(p; \sigma) = c_\psi [\psi^*(Q; \sigma)]^\omega = (1 - \sigma)^{-\eta} h_\psi(p).$$

With w_s , the success probabilities, and $\hat{V}$ fixed, the expected gain $G(p)$ is fixed, and (15) gives $h_X(p; \sigma) = (1 - \sigma)^{-1/\gamma} h_X(p)$. Differentiating each at $\sigma = 0$,

$$\frac{\partial h_\psi}{\partial \sigma} = \eta h_\psi(p), \quad \frac{\partial h_X}{\partial \sigma} = \frac{1}{\gamma} h_X(p),$$

so the share of the additional R&D labor that goes to process efficiency is

$$\varsigma(p) = \frac{\partial h_\psi / \partial \sigma}{\partial h_\psi / \partial \sigma + \partial h_X / \partial \sigma} = \frac{\gamma \eta \chi(p)}{\gamma \eta \chi(p) + 1 - \chi(p)}.$$

Differentiating, $\partial \varsigma / \partial \chi = \gamma \eta / [\gamma \eta \chi + 1 - \chi]^2 > 0$, so ς is strictly increasing in χ . By Proposition 2, $\chi(p) \rightarrow 1$ under Assumption 2, so $\varsigma(p) \rightarrow 1$. $\square$

Proof of Proposition 4 (Subsidies to large firms discourage innovation by smaller firms).

The subsidized firms raise their process efficiency, so their lines become harder to take. Consider a smaller firm, whose own subsidy is unchanged.

Part (a). In (11), the firm's chance of beating a subsidized incumbent, $\Lambda(A(\psi(p), \psi'))$, falls, and its chance against any other incumbent is unchanged. With the distribution of firms fixed, Φ_A and the weights in S are unchanged, so $B(\psi(p))$ falls, strictly if subsidized firms hold a positive share of active lines and $\kappa < \infty$. An entrant is a firm with an empty portfolio, so the same argument shows that $B_e = B(0)$ falls.

Part (b). In (13), the term for inactive lines is unchanged, and each capture term is the same probability times the gain $\hat{V}(p \oplus (1 + \lambda)x) - \hat{V}(p)$, which is positive by Lemma C.2. Hence $G(p)$ falls, and since w_s is fixed, (15) implies that $X^*(p)$ falls. For

an entrant, each gain is the value $\hat{V}$ of a one-line portfolio, which is positive, so $G(\emptyset)$ falls and X_e falls by (16). $\square$

Derivation of the growth rate (19)

Let $T \equiv \int_{\mathcal{N}} q_j dj = \bar{q} \Phi_A$ be the aggregate quality of active lines. On the balanced growth path Φ_A is constant, so $g = \dot{T}/T$. T rises when active lines are improved or inactive lines re-enter the economy, and it falls when lines exit. An active line held by a firm with process efficiency ψ' is displaced, and hence improved by the factor $1 + \lambda$, at rate $\tau(\psi')$, which adds $\lambda \bar{q} \Phi_A \int x \tau(\psi') dS(x, \psi')$ per unit of time. Inactive lines receive attempts at rate $\bar{\tau}$, and each re-enters the economy at $(1 + \lambda)\bar{q}$ after its first attempt, which adds $(1 + \lambda) \bar{q} \bar{\tau} \Phi_I$. Lines exit at rate φ regardless of their quality, which removes $\varphi \bar{q} \Phi_A \int x dS = \varphi \bar{q} \Phi_A$. Stationarity of Φ_A requires $\bar{\tau} \Phi_I = \varphi \Phi_A$. Hence

$$\dot{T} = \bar{q} \Phi_A \left[\lambda \int x \tau(\psi') dS(x, \psi') + (1 + \lambda)\varphi - \varphi \right],$$

and dividing by T gives (19).

Growth effects of a subsidy to large firms

Direct innovation (positive). The subsidy lowers the effective wage at subsidized firms and raises $X^*(p)$ through (15). By (12), this raises the displacement rate of every line they attack, hence g in (19) rises.

Blocking (negative). The subsidy raises ψ at subsidized firms. By Lemma C.1, the displacement rate of their lines falls, which lowers the quality-weighted displacement rate in (19).

Crowding out (negative). By Proposition 3, the additional demand for labor at large subsidized firms is concentrated in process R&D. Aggregate demand for skilled labor rises, and with supply fixed at L^S the wage w_s must rise to clear (18). The higher wage lowers $X^*(p)$ at unsubsidized firms and lowers X_e , which reduces displacement rates and growth.

D. STATIC VS. DYNAMIC CHOICE OF PROCESS EFFICIENCY

The process efficiency (5) solves (4), which lets ψ affect only current profit. Here I let the firm take into account that ψ also affects its chances of winning and keeping lines, and show that the static formulation understates process efficiency at every portfolio.³

D.1. *Dynamic value function*

Let $\hat{V}(p; \psi)$ denote the value, normalized by $\bar{q}$, of a firm with portfolio p that maintains process efficiency ψ at the current instant. I keep continuation values at the value

3. Process efficiency is still chosen at every instant, so ψ never becomes a state variable. A formulation in which today's ψ persists and constrains tomorrow's is a different problem, which I do not solve here.

function $\hat{V}$ of (14). The question is then how a firm would set ψ today if it took into account that a higher ψ helps it win lines and keep them. The value of such a firm is

$$\begin{aligned} \hat{r} \hat{V}(p; \psi) = & \tilde{p}(1 + \psi) Q(p) - w_s c_\psi \psi^\omega - w_s h_X(p) \\ & + X^*(p) \mathcal{G}_\psi \hat{V}(p) - \tau(\psi) \mathcal{L} \hat{V}(p) - g \sum_j x_j \frac{\partial \hat{V}(p)}{\partial x_j}, \end{aligned} \quad (\text{D.1})$$

where $\mathcal{G}_\psi$ is the operator (13) with the firm's own process efficiency set to ψ , and $\mathcal{L}$ is the loss operator of Appendix C. Besides current profit, the firm's own ψ enters (D.1) through the expected gain from an attempt,

$$\begin{aligned} \mathcal{G}_\psi \hat{V}(p) = & \Phi_I [\hat{V}(p \oplus (1 + \lambda)) - \hat{V}(p)] \\ & + \Phi_A \int \Lambda((1 + \lambda)(1 + \psi) - (1 + \psi')) [\hat{V}(p \oplus (1 + \lambda)x) - \hat{V}(p)] dS(x, \psi'), \end{aligned}$$

which rises with ψ because a higher ψ raises the firm's chance of winning every contest and each gain is positive by Lemma C.2, and through the displacement rate $\tau(\psi)$ in (12), which falls with ψ by Lemma C.1.

D.2. Dynamic first-order condition

Differentiating (D.1) with respect to ψ , with continuation values fixed and the induced adjustment of X^* removed by the envelope theorem, gives the first-order condition

$$\underbrace{\omega w_s c_\psi \psi^{\omega-1}}_{L(\psi)} = \underbrace{\tilde{p} Q(p) + X^*(p) \frac{\partial \mathcal{G}_\psi \hat{V}(p)}{\partial \psi}}_{\text{Attack term } \mathcal{A}(p; \psi)} + \underbrace{|\tau'(\psi)| \mathcal{L} \hat{V}(p)}_{\text{Defense term } \mathcal{D}(p; \psi)}. \quad (\text{D.2})$$

$\underbrace{\hspace{15em}}_{R(p; \psi)}$

The left-hand side is the marginal cost of process efficiency. On the right, $\tilde{p} Q(p)$ is the static benefit, the attack term is the value of winning more contests, and the defense term is the value of losing lines less often. Setting $\mathcal{A} = \mathcal{D} = 0$ gives back (5).

PROPOSITION D.1 (The static formulation is a lower bound). *Equation (D.2) has at least one solution, and every solution satisfies $\psi_{\text{dyn}}(p) > \psi^*(Q(p))$ for every nonempty portfolio p .*

Proof. For existence, note that $L(0) = 0$ while $R(p; 0) \geq \tilde{p} Q(p) > 0$. The slope of the logistic function is bounded, $\Lambda' \leq \kappa/4$, so

$$\frac{\partial \mathcal{G}_\psi \hat{V}(p)}{\partial \psi} \leq \frac{(1 + \lambda)\kappa}{4} \int [\hat{V}(p \oplus (1 + \lambda)x) - \hat{V}(p)] dS(x, \psi') \quad \text{and} \quad |\tau'(\psi)| \leq \frac{\bar{\tau} \kappa}{4}.$$

The integral is finite by condition (vi) of Definition 1, and $\mathcal{L} \hat{V}(p)$ is finite, so $R(p; \cdot)$ is bounded above. Since L is continuous and unbounded and $R(p; \cdot)$ is continuous and bounded, $L - R$ changes sign, and a root exists by the intermediate value theorem.

For the bound, both correction terms are strictly positive at every ψ . In the attack term, $X^*(p) > 0$ because $G(p) > 0$ by Lemma C.2, $\Lambda' > 0$, and each increment is positive by the same lemma. In the defense term, $|\tau'(\psi)| > 0$, and $\mathcal{L}\hat{V}(p) > 0$ for a nonempty portfolio by the lemma. Hence $R(p; \psi) > \tilde{p}Q(p)$ for every ψ . Let ψ_{dyn} be any root. Then

$$L(\psi_{\text{dyn}}) = R(p; \psi_{\text{dyn}}) > \tilde{p}Q(p) = L(\psi^*(Q(p))),$$

where the last equality is the static first-order condition. For $\omega > 1$, L is strictly increasing on $\psi > 0$, so $\psi_{\text{dyn}} > \psi^*(Q(p))$. $\square$

Since Λ' first rises and then falls, $R(p; \cdot)$ need not be monotone, so there may be several roots, all of which exceed the static solution.

Both corrections are small for a firm that wins or loses almost every contest with certainty, since Λ' is then close to zero. The attack motive applies to entrants as well, who would choose $\psi > 0$ to raise B_e . This works against blocking, so the implications below concern incumbents, and the net effect on entry is ambiguous.

D.3. Implications for the results of the paper

1. **Process R&D is understated.** Process R&D labor $h_\psi = c_\psi \psi^\omega$ is increasing in ψ , which is larger under the dynamic formulation at every portfolio.
2. **Blocking is understated.** The advantage $A(\psi'', \psi) = (1 + \lambda)(1 + \psi'') - (1 + \psi)$ falls with the defender's ψ . Holding challengers' process efficiency fixed, a higher ψ at incumbents therefore raises the probability $1 - \Lambda(A(\psi'', \psi))$ that an attempt fails.
3. **The welfare gains from reallocation are understated.** To the extent that the gains in Section 5 operate through these two channels, the static formulation understates them.

E. BREAKTHROUGH PATENT CONSTRUCTION

This appendix describes the breakthrough-patent measure used in Appendix A and Section 3.2. I adopt the text-based inventiveness score of Kelly et al. (2021), which is based on the content of each patent's claims rather than its citations.

The score asks whether the text of a patent resembles later patents more than earlier ones.

I construct the score in four steps. First, I tokenize the claims of each patent with the Jieba segmenter, the standard tool for Chinese word segmentation. Second, I filter the vocabulary. I keep the 435,005 tokens that appear in at least 10 patents and ask the DeepSeek large language model to classify each of them as technical or non-technical. The 199,782 technical tokens form the final vocabulary. Without this filter, common characters and grammatical particles would dominate the token counts.

Third, I represent each patent by its term frequency–inverse document frequency (TF-IDF) vector over this vocabulary, namely the count of each token in the claims weighted by the log of the inverse share of patents that contain it. I compute the weights from patents filed in the ten years before the focal patent, so that they use only information available at the filing date.

Fourth, for each patent i filed in year t_i , I compute the average cosine similarity between its vector and those of patents filed in the ten years before t_i , and the same average for the ten years after t_i . Both averages are taken within the patent’s IPC section, so that similarity reflects technological proximity rather than accidental overlap between unrelated fields. The inventiveness score is the ratio of the two,

$$\text{inv}(i) = \frac{\text{sim}_{\text{fwd}}(i)}{\text{sim}_{\text{back}}(i)},$$

so a patent with $\text{inv}(i) > 1$ is closer to later filings than to earlier ones. The measure differs from that of Kelly et al. (2021) in three ways: the Chinese tokenization, the filtering of the vocabulary by a language model, and the restriction of comparisons to the same IPC section.

To obtain a breakthrough indicator, I take the log of $\text{inv}(i)$ and subtract its mean in each year. The demeaning removes an upward trend in raw inventiveness that arises mechanically as the pool of comparison patents grows. I then classify the top 10% of the demeaned distribution as breakthrough patents.

F. REDUCED-FORM ROBUSTNESS

This appendix checks the results of Table 1 in five ways: alternative cutoffs for subsidy concentration, breakthrough patents as the dependent variable, two-way clustering, restrictions on the sample, and a pre-trend test of the Bartik instrument.

F.1. *Alternative cutoffs*

The concentration measure Conc_{jt}^p defines large firms as those above the 90th percentile of employment in the cell. Table F.1 reruns the column 3 specification with $p \in \{50, 75, 95, 99\}$. The small-firm coefficient is negative and significant at every cutoff and under both estimators. At $p = 50$ and $p = 75$ its magnitude is close to the baseline. At $p = 95$ and $p = 99$ both the small- and the large-firm coefficients rise in absolute value, as the group of large firms becomes small and its subsidy share volatile.

F.2. *Breakthrough patents*

I next replace the dependent variable with $\log(1 + \text{BT}_{jt,g})$, where BT counts breakthrough patents, the top decile of the inventiveness measure of Appendix E. If the crowding-out result came from strategic filing, it should weaken for these patents.

Table F.2 reruns the four columns of Table 1 with this dependent variable. The small-firm coefficient is negative and significant in OLS in all four columns. Under IV

TABLE F.1
Alternative concentration cutoffs for the column 3 specification

	Concentration cutoff p				
	50	75	90	95	99
<i>Panel A: OLS</i>					
$\text{Conc}^p \times \text{Small}$	-0.889*** (0.319)	-0.738*** (0.172)	-0.682*** (0.171)	-0.713*** (0.196)	-0.951*** (0.284)
$\text{Conc}^p \times \text{Large}$	0.395 (0.319)	0.063 (0.213)	0.240 (0.189)	0.375** (0.183)	0.717*** (0.214)
<i>Panel B: Bartik IV</i>					
$\text{Conc}^p \times \text{Small}$	-8.394*** (2.889)	-3.856*** (1.179)	-6.703*** (1.475)	-11.101*** (2.255)	-25.445*** (5.428)
$\text{Conc}^p \times \text{Large}$	1.060 (1.862)	-0.045 (0.887)	1.217 (1.028)	3.865*** (1.379)	16.758*** (5.069)
First-stage F (S/L)	56 / 62	119 / 123	95 / 89	78 / 74	60 / 64
Observations (OLS / IV)	5,664 / 4,077	5,664 / 4,077	5,664 / 4,077	5,664 / 4,077	5,664 / 4,077

Notes: Each column reports the column 3 specification of Table 1 at a different concentration cutoff p . The dependent variable is $\log(1 + \text{Patents}_{jt,g})$. Controls match Table 1, column 3, with Conc^p and RevConc^p replaced at the corresponding percentile. Standard errors, clustered at the 3-digit industry level, are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

it remains negative, with magnitudes close to the baseline but wider standard errors. The large-firm coefficient is small and not significantly different from zero in columns 2–4 under both estimators.

TABLE F.2
Breakthrough patents as the dependent variable

	(1)	(2)	(3)	(4)
<i>Panel A: OLS</i>				
$\text{Conc}^{90} \times \text{Small}$	-2.021*** (0.119)	-0.542*** (0.137)	-0.546*** (0.137)	-0.371*** (0.075)
$\text{Conc}^{90} \times \text{Large}$	2.387*** (0.218)	0.258 (0.262)	0.265 (0.256)	0.123 (0.078)
<i>Panel B: Bartik IV</i>				
$\text{Conc}^{90} \times \text{Small}$	-2.214*** (0.170)	-2.142* (1.121)	-2.174* (1.130)	-1.340** (0.611)
$\text{Conc}^{90} \times \text{Large}$	2.264*** (0.183)	-1.105 (1.103)	-1.358 (1.144)	-0.618 (0.694)
First-stage F (S/L)	7,374 / 6,972	95 / 92	95 / 89	195 / 175
Observations (OLS / IV)	5,664 / 4,077	5,664 / 4,077	5,664 / 4,077	13,431 / 9,900

Notes: Same specification as Table 1, with the dependent variable replaced by $\log(1 + \text{BT}_{jt,g})$, where BT is the count of breakthrough patents (top-decile inventiveness, Appendix E). Standard errors, clustered at the 3-digit industry level, are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F.3. *Two-way clustering*

The baseline specification clusters standard errors by 3-digit industry. If the residuals are also correlated across industries within a year, for example because of common shocks to subsidy allocation, these standard errors would be understated. Table F.3 reruns Table 1 with two-way clustering by industry and year.

The point estimates are the same as in Table 1. The IV standard errors in columns 2–4 are larger, but the small-firm coefficient remains significant at the 1% level.

TABLE F.3
Two-way clustering by industry and year

	(1)	(2)	(3)	(4)
<i>Panel A: OLS</i>				
$\text{Conc}^{90} \times \text{Small}$	−3.584*** (0.182)	−0.675*** (0.164)	−0.682*** (0.153)	−0.557*** (0.098)
$\text{Conc}^{90} \times \text{Large}$	2.802*** (0.121)	0.242 (0.219)	0.240 (0.208)	0.215* (0.101)
<i>Panel B: Bartik IV</i>				
$\text{Conc}^{90} \times \text{Small}$	−4.168*** (0.233)	−6.653*** (1.736)	−6.703*** (1.669)	−4.700*** (1.070)
$\text{Conc}^{90} \times \text{Large}$	3.350*** (0.187)	1.615 (0.951)	1.217 (0.954)	1.167 (0.888)

Notes: Same specification as Table 1, with standard errors using two-way clustering at the 3-digit industry and year level (Cameron et al., 2011). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F.4. *Sample splits*

Table F.4 reruns the column 3 specification on four restricted samples. I drop, in turn, the top 1% of firms by employment, the top 1% by subsidy receipts, firms whose reported subsidies exceed their revenue, and firms designated as technology SMEs. These restricted samples exclude, respectively, the most extreme firms, likely cases of misreporting, and firms favored through a separate program.

The small-firm coefficient is negative and significant in every sample, between −0.65 and −1.19 in OLS and between −5.53 and −7.33 under IV. Dropping the largest subsidy recipients gives the smallest IV estimate in absolute value, −5.53, and the estimates in the other restricted samples are close to the baseline.

F.5. *Pre-trend test*

The Bartik instrument B_{jt} uses persistent differences across provinces in their bias toward large firms, weighted by the 2008 provincial employment shares of each industry. If these shares were correlated with earlier trends in patenting that differ by size group, the instrument would be invalid. Following Jaeger et al. (2018), I regress the change in patenting within a cell, $\Delta y_{jt,g} = y_{jt,g} - y_{j,t-2,g}$, on $B_{jt} \times \text{Small}$ and $B_{jt} \times \text{Large}$.

TABLE F.4
Sample splits for the column 3 specification

Sample	OLS		Bartik IV	
	Small	Large	Small	Large
Baseline (Tab. 1, col 3)	−0.682*** (0.171)	0.240 (0.189)	−6.703*** (1.475)	1.217 (1.028)
Drop top-1% emp.	−0.792*** (0.172)	0.262 (0.164)	−7.327*** (1.666)	2.303** (1.121)
Drop top-1% subsidy	−1.194*** (0.223)	−0.081 (0.222)	−5.533*** (1.095)	1.402 (0.865)
Drop subsidy > revenue	−0.655*** (0.172)	0.276 (0.199)	−6.760*** (1.505)	1.357 (1.090)
Drop tech-SME	−0.678*** (0.172)	0.128 (0.186)	−6.058*** (1.516)	1.111 (1.130)

Notes: Each row applies a firm-level filter to the working sample, then re-runs the column 3 specification of Table 1. Standard errors clustered at the 3-digit industry level in parentheses.
 * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The coefficient on the small-firm interaction is 0.091 ($p = 0.12$), not significantly different from zero, so the small-firm estimate in Table 1 does not reflect earlier differences in patenting trends across industries. The coefficient on the large-firm interaction is 0.155 ($p < 0.01$). It is small relative to the second-stage IV estimate but significant, so the large-firm IV estimate should be read with caution.

G. STANDARD ERRORS

This appendix describes the standard errors of the SMM estimates in Table 2.

Sandwich estimator. I use the SMM variance estimator of Gouriéroux and Monfort (1996), as in Garcia-Macia et al. (2019),

$$\widehat{\text{Var}}(\hat{\theta}) = (G'WG)^{-1} G'W \hat{\Omega} WG (G'WG)^{-1}, \quad (\text{G.1})$$

where $G \in \mathbb{R}^{7 \times 6}$ is the Jacobian of the model moments with respect to the parameters at $\hat{\theta}$, $W \in \mathbb{R}^{7 \times 7}$ is the weighting matrix in (24), and $\hat{\Omega} \in \mathbb{R}^{7 \times 7}$ is the sampling covariance of the data moments. Since the model moments are computed without simulation, the simulation-noise term of the general formula is zero.

Bootstrap of $\hat{\Omega}$. I estimate the covariance of the moments with a firm-cluster bootstrap of the Chinese Tax Survey microdata. In each of $B = 500$ replicates, I draw $n = 52,091$ firms with replacement from the working sample and keep every observation of each drawn firm, so a firm drawn twice appears twice with its full history. I then recompute the inverse-probability weights that match the firm-size distribution of the Economic Census, using the bin shares of the bootstrap sample, and recompute the moments as in Section 3. Resampling firms preserves the serial correlation within

firms on which the slope moments depend. For each moment, $\hat{\Omega}$ uses the variance of the estimator that produces the corresponding data target, and (G.1) carries this uncertainty to the six parameters.

Numerical Jacobian. I compute G by central differences in the logs of the parameters,

$$(G_{\log})_{ik} = \left. \frac{\partial m_i}{\partial \log \hat{\theta}_k} \right|_{\hat{\theta}} \approx \frac{m_i(\hat{\theta} \cdot e^{+h} e_k) - m_i(\hat{\theta} \cdot e^{-h} e_k)}{2h}, \quad (\text{G.2})$$

with $h = 10^{-2}$, a one percent perturbation of each parameter. Differentiating in logs allows me to apply the same relative step to parameters of different magnitudes while keeping every perturbed parameter positive. It also multiplies each column of the raw Jacobian by the corresponding parameter, $G_{\log} = G_{\text{raw}} \cdot \text{diag}(\hat{\theta})$, which puts the columns on a comparable scale and improves the conditioning of $G'WG$.

Reporting scale. Applying (G.1) with $G_{\log}$ gives $\widehat{\text{Var}}(\log \hat{\theta})$. Table 2 reports standard errors on the log scale, $\text{SE}(\log \hat{\theta}_k)$, which are unit-free and can be read as relative standard errors. Confidence intervals are then multiplicative,

$$[\hat{\theta}_k \exp(-1.96 \text{SE}(\log \hat{\theta}_k)), \hat{\theta}_k \exp(+1.96 \text{SE}(\log \hat{\theta}_k))].$$

Unlike the symmetric interval $\hat{\theta}_k \pm 1.96 \hat{\theta}_k \text{SE}(\log \hat{\theta}_k)$, which has a negative lower bound whenever $\text{SE}(\log \hat{\theta}_k) > 1/1.96 \approx 0.51$, they respect the positivity of the parameters. Standard errors on the original scale follow from the delta method,

$$\widehat{\text{Var}}(\hat{\theta}) = \text{diag}(\hat{\theta}) \widehat{\text{Var}}(\log \hat{\theta}) \text{diag}(\hat{\theta}),$$

so that $\text{SE}(\hat{\theta}_k) = \hat{\theta}_k \cdot \text{SE}(\log \hat{\theta}_k)$.

H. SUPPLEMENTARY ESTIMATION RESULTS

H.1. The Pat/R&D profile

Figure H.1 plots the size profile of Pat/R&D in the model as the cost scale c_ψ varies, with the other parameters held at their estimates. At $\hat{c}_\psi = 0.32$, the profile falls from 0.24 at $n = 1$ to 0.04 at $n = 15$. As c_ψ rises the profile flattens, and in the Klette–Kortum limit it is constant in n , as the argument of Section 4.3 implies.

H.2. Re-estimation without process efficiency

To shut down the process efficiency mechanism, I re-estimate the model with $c_\psi \rightarrow \infty$, minimizing the SMM objective over $(\theta, \lambda, H, \alpha_e)$. The fitted slope of $\log(\text{Pat/R\&D})$ is +0.12 rather than -0.56 , and the other three moments in Table H.1 are further from the data than under the full model.

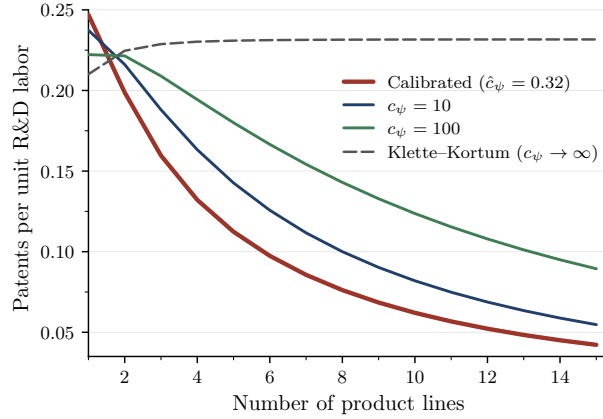


FIGURE H.1
Patents per unit of R&D labor as c_ψ varies

Notes: The figure plots the model-implied schedule $\text{Pat}(n)/\text{R\&D}(n)$ against the number of product lines n at four values of the process-efficiency cost scale c_ψ : 0.03, the estimated $\hat{c}_\psi = 0.32$, 30, and the Klette–Kortum limit $c_\psi \rightarrow \infty$. The remaining parameters ($\omega, \theta, \lambda, H, \alpha_e$) are held at their estimated values.

TABLE H.1
Model fit: full versus restricted model

Moment	Data	Full model	Restricted
Aggregate growth	0.0222	0.0231	0.0407
IQR(log Emp)	1.39	1.08	0.60
log(R&D/Rev) slope	−0.61	−0.52	−1.24
log(Pat/R&D) slope	−0.56	−0.55	+0.12

Notes: The restricted model shuts down process efficiency by imposing $c_\psi \rightarrow \infty$ and re-estimates the remaining parameters against the benchmark SMM objective. The restricted estimates are $\hat{\theta} = 0.36$, $\hat{\lambda} = 0.55$, $\hat{H} = 1.00$, and $\hat{\alpha}_e = 2.78$. Slopes are coefficients from regressions of the indicated log ratio on log employment.

H.3. The crowding-out coefficient in the model

As a check on the general-equilibrium channels of the model, I ask whether the calibrated model reproduces the crowding-out coefficient of the preferred specification in Table 1. With all parameters at their estimates, I impose nineteen subsidy schedules that raise the rate at the top of the size distribution while holding the small-firm rate fixed, and I solve for the equilibrium under each. I then regress the gap in log patents per firm between small and medium firms, $\log(\text{Pat}_S^{\text{avg}}) - \log(\text{Pat}_M^{\text{avg}})$, on Conc^{90} across the nineteen equilibria, treating the model as a single industry-year.⁴ The small-firm

4. The baseline gives all firms a uniform 5% discount on the cost of R&D labor. Each step then adds 2 percentage points to the discount of firms above p99 and 1 point to that of firms in p90–p99, roughly in line with the data, and leaves the other firms unchanged. The nineteen schedules generate values of Conc^{90} between 0.28 and 0.80. Small firms are those below p47 of employment and medium firms those between p47 and p86, which match the 35- and 395-employee cutoffs in the data. The dependent variable is X^*B , the rate of successful product innovations per firm, which is the model analog of patents per firm.

coefficient in the model is -0.38 ($R^2 = 0.97$), slightly over half of the OLS estimate in column 3 (-0.68). Two features of the model may account for the gap. Its firm-size distribution is more compressed than in the data, which dampens the responses of the wage and of blocking to subsidy concentration, and it omits channels such as credit constraints that bind hardest on small firms.

H.4. *The calibrated economy*

Figure H.2 plots two features of the economy at the estimates in Table 3 of the main text. Panel (a) shows that the share of R&D labor devoted to process efficiency rises from 13% at single-line firms to 67% at firms with eleven lines. Panel (b) shows that the probability that an attempt against a firm with n lines fails rises from close to zero for single-line firms to above 0.12 for firms with eleven or more lines.

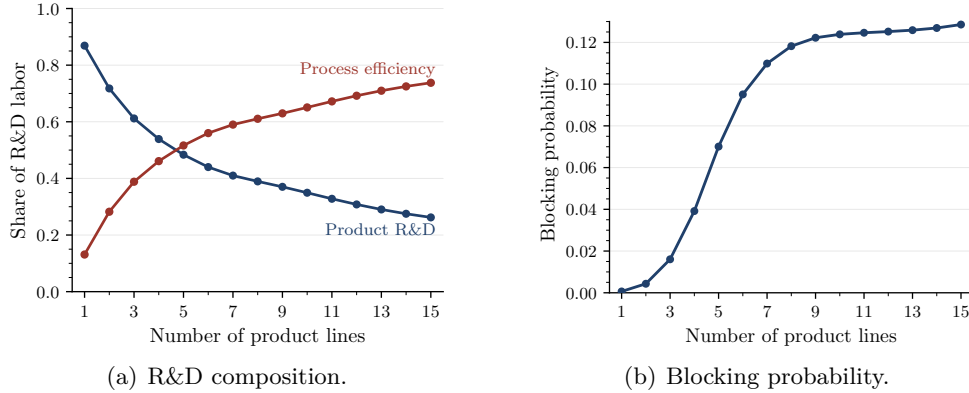


FIGURE H.2
Features of the calibrated economy

Notes: Panel (a) plots the composition of R&D labor against the number of product lines n . Panel (b) plots the probability that an attempted innovation against a size- n incumbent fails. Both panels are computed at the parameter estimates of Table 3 of the main text.

H.5. *Consumption-equivalent variation*

I evaluate welfare on the balanced growth path with the household's CRRA preferences and initial consumption net of the tax, $C_0 = \hat{Y} - T$. Let (C_0, g_0) denote baseline consumption and growth, and (C_1, g_1) consumption and growth under the alternative policy. The CEV ξ solves $W(\xi C_0, g_0) = W(C_1, g_1)$, so that

$$\xi = \frac{C_1}{C_0} \left[\frac{\rho - (1 - \vartheta) g_0}{\rho - (1 - \vartheta) g_1} \right]^{1/(1-\vartheta)}. \quad (\text{H.1})$$

I report $\xi - 1$ in percent.

I. COMPUTATIONAL ALGORITHM

The algorithm has two nested loops. The inner loop takes the skilled wage w_s as given and computes the remaining objects of the balanced growth path. The outer loop bisects on w_s to clear the market for skilled labor (18).

The inner loop proceeds as follows.

1. Initialize the aggregate attempt rate $\bar{\tau}$, the share of active lines Φ_A , average process efficiency $\bar{\Psi}$, and the distribution of firms $\mu(n)$ on a grid $n \in \{1, \dots, N_{\max}\}$.
2. Compute the profit coefficient $\tilde{p}$ from (B.2).
3. Compute process efficiency $\psi^*(n)$ from (5). Under a subsidy schedule, map firm sizes to employment, assign the subsidy rates $\sigma(n)$, and use the effective wage $(1 - \sigma(n))w_s$ in place of w_s .
4. Compute the success probabilities $B(n)$ and displacement rates $\tau(n)$ from (11) and (12).
5. Solve for the value function $\hat{V}(n)$ by policy iteration, using the Thomas algorithm for the resulting tridiagonal system.
6. Obtain the attempt rates $X^*(n)$ from the first-order condition (15).
7. Compute the entry rate X_e from the free-entry condition (16), by bisection when $\alpha_e > 0$ and as residual labor when $\alpha_e = 0$.
8. Update $\mu(n)$ by solving the flow equations for the number of lines, again a tridiagonal system, with the entry source at $n = 1$.
9. Update Φ_A , $\bar{\Psi}$, $\bar{\tau}$, and the mass of firms $F = \Phi_A/\bar{n}$.
10. If μ has not converged, set $\mu \leftarrow (1 - \delta)\mu + \delta\mu_{\text{new}}$ with $\delta = 0.4$ and return to step 2.

To save time, I set $N_{\max} = 80$ and raise it to $N_{\max} = 640$ only when more than 0.5% of the mass of μ lies above $0.85N_{\max}$. The convergence tolerance is 10^{-7} in the inner loop and 10^{-6} in the outer bisection.

J. ROBUSTNESS OF THE QUANTITATIVE RESULTS

This appendix reports the parameter estimates and welfare gains for the robustness exercises of Section 5.4.

J.1. *Parameter estimates*

Table J.1 reports the estimates under the five specifications. All of them stay within about one standard error of the benchmark. The entry-cost parameters H and α_e move the most, since they partly substitute for each other in the free-entry condition.

TABLE J.1
Parameter estimates across robustness specifications

Parameter	Description	Benchmark	+ Yield	+ Process	$\gamma = 0.4$	$\gamma = 0.6$
ω	Process efficiency cost convexity	3.279	3.202	3.223	3.276	3.198
c_ψ	Process efficiency cost scale	0.323	0.260	0.339	0.361	0.398
θ	Incumbent innovation capacity	0.149	0.145	0.149	0.149	0.166
λ	Quality step size	1.079	1.177	1.072	1.047	1.005
α_e	Entry-cost convexity	3.000	2.615	3.000	2.914	3.000
H	Entry-cost level	0.527	0.328	0.534	0.476	0.598

Notes: Each column re-estimates the model under a different specification. “+ Yield” keeps the seven benchmark moments and adds an eighth targeted moment, the ratio of patents per subsidy yuan for subsidized firms in the p50–p90 size bin relative to firms below p50, computed within 4-digit industry $\times$ year cells. “+ Process” instead adds the share of large product innovators that also conduct process innovation (87.1%) as the eighth moment (Section 5.4). The γ columns fix the R&D production-function curvature and re-estimate the remaining six parameters against the benchmark moments. The externally calibrated parameters remain unchanged.

J.2. Welfare outcomes

Table J.2 reports the welfare gains under each specification for the three counterfactuals of Section 5. The conclusions do not change. Removing the existing subsidies raises welfare by about 1% in consumption-equivalent terms, the budget-neutral optimum by 10–14%, and the variable-budget optimum by 18–22%.

TABLE J.2
Welfare gains across robustness specifications (CEV $\xi - 1$, %)

Counterfactual	Benchmark	+ Yield	+ Process	$\gamma = 0.4$	$\gamma = 0.6$
No subsidy	1.02	1.11	1.03	1.08	0.97
Budget-neutral optimum	10.73	12.33	10.69	13.64	12.59
Variable-budget optimum	18.20	21.54	18.04	21.39	22.40

Notes: Each entry is the consumption-equivalent welfare gain (CEV $\xi - 1$ in percent) relative to the baseline calibration of the corresponding specification, computed at $\rho = 0.020$ and $\vartheta = 2$. “+ Yield” adds the ratio of patents per subsidy yuan for subsidized firms in the p50–p90 size bin relative to firms below p50 as an eighth targeted moment, and “+ Process” adds the share of large product innovators that also conduct process innovation (Section 5.4). The γ columns fix the R&D production-function curvature and re-estimate the remaining parameters. The benchmark column reproduces Panel D of Table 5.

The variable-budget gain varies the most, from 18.0% to 22.4%, because this counterfactual is the most sensitive to the quality step λ and to the curvature γ of the R&D production function.

K. FIRM TYPE HETEROGENEITY EXTENSION

This appendix adds permanent differences in innovation productivity across firms to the model, as in [Acemoglu et al. \(2018\)](#). Larger firms’ reliance on process R&D and the blocking externality then operate within each type. Types change how firms are distributed over portfolios and how often they attempt, but not their process efficiency

or the contests they face at a given portfolio. The welfare gains in Section 5 therefore understate the gains available to a planner who could also condition on type.

K.1. *Setup*

Each firm has a permanent type $\theta_f \in \{\theta^h, \theta^l\}$, with $\theta^h > \theta^l$, drawn when its entry attempt succeeds, with $\Pr(\theta = \theta^h) = \alpha$. The innovation technology becomes

$$X = \theta_f h_X^{1-\gamma},$$

which replaces (8). The rest of the model is unchanged.

K.2. *Process efficiency and blocking*

The process-efficiency problem (4) does not involve θ . Hence the optimal choice

$$\psi^*(Q) = \left(\frac{\tilde{p} Q}{\omega w_s c_\psi} \right)^{1/(\omega-1)},$$

the flow profit $\pi(Q) = \tilde{p} Q + \hat{D} Q^\eta$, and process R&D labor $h_\psi = c_\psi [\psi^*(Q)]^\omega$ depend on a firm's portfolio only through its aggregate quality.

Blocking has the same structure. The advantage $A(\psi, \psi') = (1+\lambda)(1+\psi) - (1+\psi')$ depends on the process efficiencies of the challenger and the incumbent but not on their types. The success probability keeps the form of (11), and the displacement rate (12) becomes

$$\tau(\psi) = X_e \Lambda(A(0, \psi)) + \sum_{k \in \{h, l\}} \int X^k(p'') \Lambda(A(\psi(p''), \psi)) dM_k(p''),$$

where M_k is the measure of type- k firms over portfolios. Proposition 1 holds because its proof requires only that ψ^* be increasing in Q , and Proposition 2 holds within each type.

K.3. *Innovation and the distribution of firms*

Types enter through the dynamic problem. Let $\hat{V}_k(p)$ denote the value of a type- k firm with portfolio p , and $G_k(p) \equiv \mathcal{G} \hat{V}_k(p)$ its expected gain from an attempt, with $\mathcal{G}$ given by (13). The value function (14) becomes

$$\hat{r} \hat{V}_k(p) = \max_{X \geq 0} \left\{ \pi(Q(p)) - w_s \left[\frac{X}{\theta_k} \right]^{\frac{1}{1-\gamma}} + X G_k(p) \right\} - \tau(\psi(p)) \mathcal{L} \hat{V}_k(p) - g \sum_j x_j \frac{\partial \hat{V}_k(p)}{\partial x_j}$$

for $k \in \{h, l\}$, with the same discount rate $\hat{r} = \rho + (\vartheta - 1)g + \varphi$ as in the baseline. The two value functions differ only in the cost of product R&D. The first-order condition gives

$$X^k(p) = \theta_k^{1/\gamma} \left[\frac{(1-\gamma) G_k(p)}{w_s} \right]^{(1-\gamma)/\gamma}.$$

Since the exponent on θ_k is $1/\gamma > 1$, a small difference in type becomes a larger difference in the attempt rate. Moreover, $\hat{V}_h(p) \geq \hat{V}_l(p)$ at every portfolio, since a high type can follow the policy of a low type at a lower R&D cost.

Types are permanent, so the stationary measure separates into two type-specific measures. Each M_k satisfies (17) with X^k in place of X^* and with the entry term multiplied by α_k , where $\alpha_h = \alpha$ and $\alpha_l = 1 - \alpha$. High types attempt more at every portfolio, so they grow faster and are overrepresented among firms with high aggregate quality.

K.4. *Equilibrium*

Types are revealed only after a successful entry attempt, so the free-entry condition (16) holds with $G(\emptyset)$ computed from the type-averaged value $\alpha \hat{V}_h + (1 - \alpha) \hat{V}_l$. The labor market sums product R&D over the two types, while process R&D depends only on aggregate quality. The aggregate attempt rate adds the incumbent attempts of both types to entry, and the growth rate keeps the form of (19), with S induced by $M_h + M_l$. Since blocking depends only on process efficiency, success and displacement probabilities aggregate across types without change. The welfare evaluation and the consumption-equivalent variation (H.1) carry over unchanged.

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